

JUnit Exercise

Module 4 - JUnit Basic Testing Exercises

Exercise 1 - Calculator.java

```
package com.cognizant.junit;

public class Calculator {
    public int add(int a,int b){ return a+b; }
    public int subtract(int a,int b){ return a-b; }
    public int multiply(int a,int b){ return a*b; }
    public int divide(int a,int b){ return a/b; }
}
```

Exercise 2 - CalculatorTest.java

```
package com.cognizant.junit;

import static org.junit.Assert.*;
import org.junit.Test;

public class CalculatorTest {
    Calculator calculator = new Calculator();

    @Test public void testAddition(){ assertEquals(15, calculator.add(10,5)); }
    @Test public void testSubtraction(){ assertEquals(5, calculator.subtract(10,5)); }
    @Test public void testMultiplication(){ assertEquals(50, calculator.multiply(10,5)); }
    @Test public void testDivision(){ assertEquals(2, calculator.divide(10,5)); }
}
```

Exercise 3 - AssertionsTest.java

```
package com.cognizant.junit;

import static org.junit.Assert.*;
import org.junit.Test;

public class AssertionsTest {
    @Test
    public void testAssertions(){
        assertEquals(5,2+3);
        assertTrue(10>5);
        assertFalse(5>10);
        assertNull(null);
        assertNotNull(new Object());
        String str="JUnit";
        assertEquals(str,str);
        assertNotSame(new String("Java"), new String("Java"));
    }
}
```

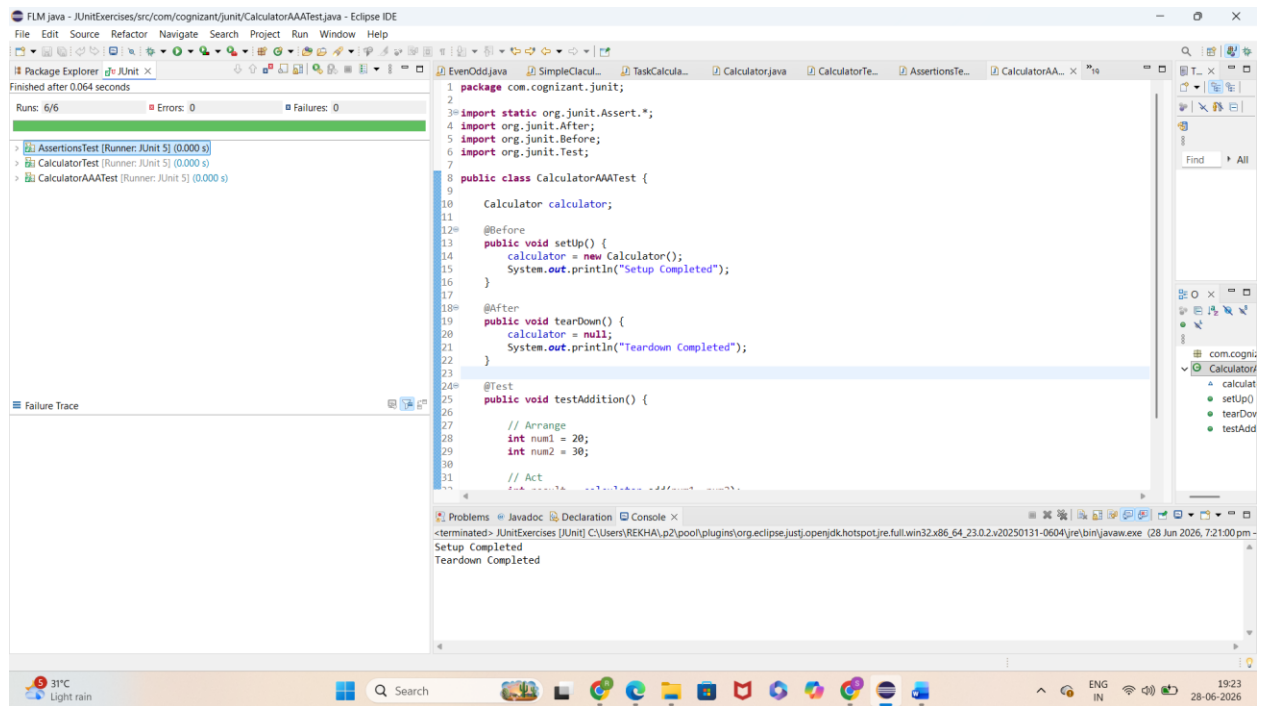
```
}  
}
```

Exercise 4 - CalculatorAAATest.java

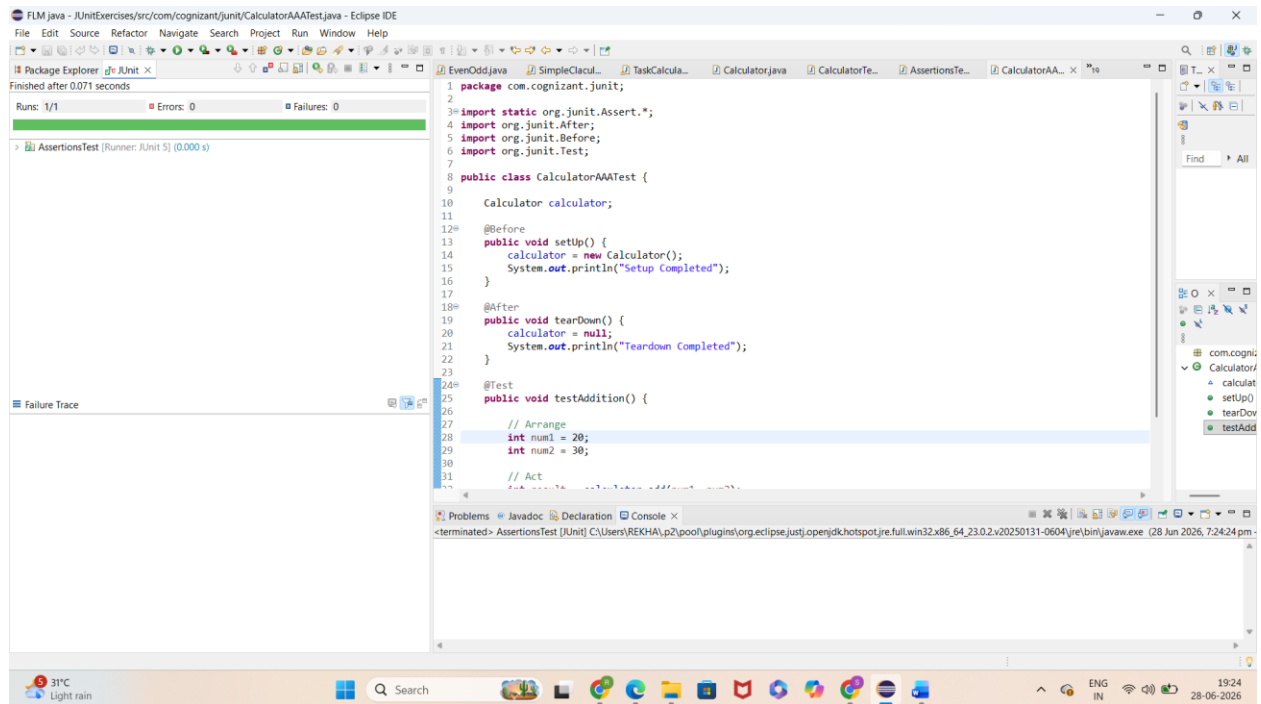
```
package com.cognizant.junit;  
  
import static org.junit.Assert.*;  
import org.junit.Before;  
import org.junit.After;  
import org.junit.Test;  
  
public class CalculatorAAATest {  
    Calculator calculator;  
    @Before  
    public void setUp(){  
        calculator=new Calculator();  
        System.out.println("Setup Completed");  
    }  
    @After  
    public void tearDown(){  
        calculator=null;  
        System.out.println("Teardown Completed");  
    }  
    @Test  
    public void testAddition(){  
        int num1=20, num2=30;  
        int result=calculator.add(num1,num2);  
        assertEquals(50,result);  
    }  
}
```

Execution Output Screenshots

Run All Tests



AssertionsTest Output



CalculatorAAATest Output

The screenshot shows the Eclipse IDE interface. The Package Explorer on the left shows the project structure with 'CalculatorAAATest' selected. The main editor displays the source code of 'CalculatorAAATest.java'. The console at the bottom shows the output of the test run, which completed successfully after 0.072 seconds. The status bar at the bottom indicates the system temperature is 31°C and it is raining.

```
package com.cognizant.junit;
import static org.junit.Assert.*;
import org.junit.After;
import org.junit.Before;
import org.junit.Test;

public class CalculatorAAATest {
    Calculator calculator;

    @Before
    public void setUp() {
        calculator = new Calculator();
        System.out.println("Setup Completed");
    }

    @After
    public void tearDown() {
        calculator = null;
        System.out.println("Teardown Completed");
    }

    @Test
    public void testAddition() {
        // Arrange
        int num1 = 20;
        int num2 = 30;

        // Act
        int result = calculator.add(num1, num2);
        // Assert
        assertEquals("Addition result is incorrect", 50, result);
    }
}
```

Finished after 0.072 seconds

Runs: 1/1 Errors: 0 Failures: 0

Failure Trace

Console

<terminated> CalculatorAAATest [JUnit] C:\Users\REKHA\p2\pool\plugins\org.eclipse.jst.openjdk.hotspot.jre.full.win32.x86_64.23.0.2\v20250131-0604\jre\bin\javaw.exe (28 Jun 2026, 7:24:45)

Setup Completed

Teardown Completed

CalculatorTest Output

The screenshot shows the Eclipse IDE interface. The Package Explorer on the left shows the project structure with 'CalculatorTest' selected. The main editor displays the source code of 'CalculatorAAATest.java'. The console at the bottom shows the output of the test run, which completed successfully after 0.069 seconds. The status bar at the bottom indicates the system temperature is 31°C and it is raining.

```
package com.cognizant.junit;
import static org.junit.Assert.*;
import org.junit.After;
import org.junit.Before;
import org.junit.Test;

public class CalculatorAAATest {
    Calculator calculator;

    @Before
    public void setUp() {
        calculator = new Calculator();
        System.out.println("Setup Completed");
    }

    @After
    public void tearDown() {
        calculator = null;
        System.out.println("Teardown Completed");
    }

    @Test
    public void testAddition() {
        // Arrange
        int num1 = 20;
        int num2 = 30;

        // Act
        int result = calculator.add(num1, num2);
        // Assert
        assertEquals("Addition result is incorrect", 50, result);
    }
}
```

Finished after 0.069 seconds

Runs: 4/4 Errors: 0 Failures: 0

Failure Trace

Console

<terminated> CalculatorTest [JUnit] C:\Users\REKHA\p2\pool\plugins\org.eclipse.jst.openjdk.hotspot.jre.full.win32.x86_64.23.0.2\v20250131-0604\jre\bin\javaw.exe (28 Jun 2026, 7:25:02 pm)